

ABG Pattern Finder

Use pH, PaCO2, and bicarbonate direction to identify the primary pattern; clinical context determines the cause.

Reference frame - pH about 7.35-7.45. - PaCO2 about 35-45 mm Hg. - HCO3 about 22-26 mEq/L. - Use the reporting laboratory range.	Direction rules - Respiratory: pH and PaCO2 move opposite. - Metabolic: pH and HCO3 move together. - A value moving in the other system may show compensation.
Clinical anchors - Ventilation problems change CO2. - Renal/GI/metabolic problems change bicarbonate or acids. - Airway and breathing instability come before worksheet interpretation.	

Reviewed 2026-07-13 - Source: MedlinePlus, U.S. National Library of Medicine - <https://medlineplus.gov/lab-tests/electrolyte-panel/>
Study aid only. Verify current references, instructor guidance, scope, orders, and facility policy.